Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 27)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
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6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

Feature	Natural systems	Linear human systems	
Energy source			
Use of materials			
Nature of wastes			
Temperature of processes			

0 1 . 2 Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

0 1 . 3 Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

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0 1 . 4 Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

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10

0 2 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 2 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 2 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

1 .....

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0 3 The Hawaiian goose (nene) was reduced by hunting, habitat loss and introduced predators to about 30 birds. A captive breeding programme started with just seven individuals and has increased the wild population to over 2000.

0 3 . 1 Explain why starting the programme with seven individuals caused problems, and describe how these problems are managed.

[4 marks]

0 3 . 2 Explain **three** reasons, other than gene pool size, why species kept in captivity may fail to breed.

[3 marks]

0 3 . 3 Describe how artificial insemination and embryo transfer increase breeding success in captive mammals.

[3 marks]

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0 4 . 1 Explain what a carbon footprint should include when calculated for a car.

[3 marks]

0 4 . 2 Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

0 4 . 3 Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

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0 5 . 1 Describe how each of the following threatens Antarctica.

[4 marks]

Global climate change

Ozone depletion

Tourism

Future mineral exploitation

0 5 . 2 Describe the main features of the Antarctic Treaty (1959) and the regulations that protect Antarctica.

[6 marks]

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0 6 . 1 Define each of the following ecological terms and give an example.

[8 marks]

Species

Ecological niche

Community

Biome

0 6 . 2 Explain the difference between an ecosystem and a biome.

[2 marks]

10

0 7 . 1 Which of the following best describes a **representative** sample? Tick one box.

[1 mark]

A A sample taken from the most convenient location ☐

B A sub-sample that accurately reflects the complete data set ☐

C The largest sample that can be collected ☐

D A sample that supports the hypothesis ☐

0 7 . 2 Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

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0 7 . 3 Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

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0 8 . 1 Explain the meaning of each of the following terms and explain why such species are given conservation priority.

[8 marks]

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EDGE species .....

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Endemic species .....

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Keystone species .....

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Flagship species .....

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0 8 . 2 Explain why the degree of population dispersal is considered when setting priorities.

[2 marks]

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0 9 Table 1 shows characteristics introduced into commercial crops from crop wild relatives (CWRs).

Table 1

Crop	Wild relative source	Characteristic introduced
Sugar cane	Wild sugar cane, Indonesia	Resistance to mosaic virus
Tomato	Single wild species	Resistance to nine major diseases
Rice / barley	Wild salt-marsh varieties	Salt tolerance
Oil palm	Wild palms, central Africa	25% higher yield
Wheat	Wild wheat	Better mycorrhizal associations / phosphate uptake

0 9 . 1 Explain why commercial crop varieties often lack the characteristics shown in Table 1.

[3 marks]

0 9 . 2 Explain what is meant by a **Vavilov centre** (centre of diversity) and why these areas are conservation priorities.

[3 marks]

0 9 . 3 Explain why a large population does not necessarily have a large gene pool, and why a small gene pool is a problem.

[3 marks]

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0 9 . 4 Give **one** wild plant species with potential as a new commercial food crop and the characteristic that makes it valuable.

[1 mark]

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10

1 0 Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

1 0 . 1 Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer ..... %

1 0 . 2 Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

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1 0 . 3 Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

1 0 . 4 Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

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Extra space

15

Write an essay on one of the following topics.

1 1 . 1 Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

OR

1 1 . 2 Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS